

Luc Illusie

Cotangent Complex and Deformations I

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Introduction

The present work is devoted to the theory of the cotangent complex on ringed topoi (a globalization of the formalism of André ([1], [2]) and Quillen ([24], [25])), and to its applications to certain problems of global infinitesimal deformation. The most interesting of these concern calculations of obstructions to the deformation of flat commutative group schemes, calculations that play a key role in the recent work of Grothendieck⁽¹⁾ and Messing⁽²⁾ on Barsotti–Tate groups.

Here, briefly, is the content of the various chapters.

Chapter I gathers the homological algebra needed for the construction and study of the properties of the cotangent complex. We introduce the notion of the homotopy sheaf of a simplicial sheaf and prove a global variant of a theorem of Whitehead [9], namely, that if a morphism $f : X \rightarrow Y$ of simplicial sheaves induces isomorphisms on homotopy sheaves in every degree, then the same is true of the morphism $\mathbb{Z}(f) : \mathbb{Z}(X) \rightarrow \mathbb{Z}(Y)$ induced on the free simplicial $\mathbb{Z}$ -modules they generate. By “inverting” the arrows that induce isomorphisms on homotopy sheaves, one defines various “derived” categories, common generalizations of Verdier’s derived categories [27] and Quillen’s homotopy categories [23]. Because of its importance for the theory of the cotangent complex, we study in particular the derived category of modules over a simplicial ring: distinguished triangles, Ext^i , and derived tensor product. The last section contains the definition of the derived functors of certain nonadditive functors on the derived category D^- of a ringed topos, and global analogues of Quillen’s shift isomorphisms ([24] 7.21).

⁽¹⁾ A. Grothendieck, *Cristaux et modules de Dieudonné*, lecture at the International Congress of Mathematicians, Nice (1970), and courses at the Collège de France, 1970–71 and 1971–72.

⁽²⁾ W. Messing, *The Crystals associated to Barsotti–Tate Groups, with Applications to Abelian Schemes*, Thesis, Princeton (1971), forthcoming.

In Chapter II, using the theory of standard resolutions and the variant of Whitehead’s theorem indicated above, we define the cotangent complex $L_{X/Y}$ of a morphism of ringed topoi $X \rightarrow Y$, and study its general functoriality and compatibility with base change. The main result, which generalizes ([25] 5.1), is that a composite $X \xrightarrow{f} Y \rightarrow Z$ of morphisms of ringed topoi gives rise to a distinguished (canonical and functorial) triangle in the derived category $D(X)$, called the “transitivity triangle”:

$$\begin{array}{ccc} & L_{X/Y} & \\ \swarrow & & \searrow \\ \mathbf{L}f^*L_{Y/Z} & \longrightarrow & L_{X/Z} \end{array}$$

In Chapter III, for a morphism of ringed topoi $X \rightarrow Y$ and an $\mathcal{O}_X$ -module I , we establish a canonical functorial isomorphism between the group of isomorphism classes of Y -extensions of X by I and the hyper-Ext group $\text{Ext}^1(L_{X/Y}, I)$. This result allows us to compare $L_{X/Y}$ with the complex constructed by Grothendieck in [17]: the latter is isomorphic in $D(X)$ to the truncation of $L_{X/Y}$ obtained by killing cohomology in degrees ≤ -2 . Combining these results with those of Chapter II, we obtain applications of the theory to questions concerning deformations of ringed topoi and morphisms of ringed topoi. We express the obstruction to a first-order flat deformation of a ringed topos as cup product with a certain fundamental class defined in terms of the transitivity triangle described above, a class that appears as a natural generalization of the classical Kodaira–Spencer invariant (cf. [16]). This result gives an affirmative solution to a conjecture of Grothendieck that lies at the origin of the present work. The rest of the chapter is devoted to computing the cotangent complex for several types of morphisms of schemes (smooth morphisms, complete intersections).

Using a variant of cotangent-complex theory for sheaves of graded algebras, in Chapter IV we treat questions concerning deformations of modules. The obstruction to a first-order flat deformation of a module is expressed as cup product with a certain fundamental class, which generalizes

the usual extension defined by the first-order jets of a module. This invariant gives rise to a theory of Chern classes for perfect complexes, presented in Chapter V after some preliminaries on derived categories of filtered complexes. In particular, the theory includes that of the characteristic polynomial of an endomorphism of a perfect complex, which resolves a question raised in (SGA 6 XIV).

Chapter VI^(*) prepares the ground for obstruction calculations concerning deformations of group schemes. It consists, for the most part, of general calculations on the cohomology of topoi associated with certain diagrams. These calculations complement those of Deligne–Saint-Donat in (SGA 4 VI), although our account is logically independent of theirs. The most striking result is the expression of Ext groups of modules over a ring as “spatial” cohomology groups of a topos defined from Eilenberg–Mac-Lane spectra. As a by-product, we obtain a remarkable explicit formula for the resolution sought by Grothendieck in (SGA 7 VII 3.5.4). The idea of the construction, which goes back to Mac-Lane⁽³⁾, was suggested to us by L. Breen, who makes essential use of this resolution in his recent work on $\text{Ext}^1(G_a, G_a)$ ⁽⁴⁾.

Chapter VII contains obstruction calculations for first-order deformations of torsors under a flat group scheme, of noncommutative flat group schemes, and of flat commutative group schemes with a scheme of operator rings. The obstructions are again described as cup products with certain fundamental classes, among which one finds (in the case of torsors) a generalization of the class constructed by Atiyah in⁽⁵⁾. The results of this

^(*) Chapters VI through VIII will be published in a second volume, which will appear shortly.

⁽³⁾ S. Mac-Lane, *Homologie des anneaux et des modules*, Colloque de Topologie algébrique, Louvain, 1956, p. 55–80.

⁽⁴⁾ L. Breen, work in preparation.

⁽⁵⁾ M. F. Atiyah, *Analytic connexions on fibre bundles*, Mexico Symposium, 1958.

chapter, which, in the case of torsors and noncommutative group schemes, complete and refine those of Demazure (SGA 3 III) and Giraud [13], had been conjectured by Grothendieck⁽⁶⁾. The method we use consists in reducing the initial problem to that of deforming a suitable diagram (for example, a group scheme will be viewed as a particular simplicial scheme), or more precisely the ringed topos of that diagram, a problem to which the general results of Chapter III apply. The formulas of Chapter VI allow the obstructions thus obtained to be written in a simpler form, involving familiar invariants such as the Lie algebra of a smooth group scheme or, more generally, the Lie complex of a flat group scheme, a complex introduced in the finite case by Mazur and Roberts in⁽⁷⁾.

Chapter VIII is independent of the results of Chapters IV through VII. It is devoted to an account of an unpublished theory of Quillen on the cohomology of formal groupoids. With a few refinements, this theory enables us to establish, in the case of local complete intersections, certain remarkable relations among the cotangent complex, the de Rham complex, and crystalline cohomology⁽⁸⁾.

The main results of Chapters I through V have been summarized in four notes in the Comptes Rendus⁽⁹⁾,^(**).

⁽⁶⁾ A. Grothendieck, letter to J. Giraud (13 June 1969), letters to L. Illusie (2 December 1969).

⁽⁷⁾ B. Mazur and L. Roberts, *Local Euler Characteristics*, Inventiones math. 9, 201–234 (1970).

⁽⁸⁾ P. Berthelot, *Cohomologie cristalline des schémas*, Thesis, forthcoming.

⁽⁹⁾ L. Illusie, *Algèbre homotopique relative*, C. R. Acad. Sc. Paris, t. 268, p. 11–14 and 206–209, *Complexe cotangent relatif d'un faisceau d'algèbres*, C. R. Acad. Sc. Paris, t. 268, p. 278–281 and 323–326.

^(**) The reader may also consult, for the work as a whole, a mimeographed summary prepared by the Faculty of Sciences at Orsay.

XIII

My gratitude goes first to A. Grothendieck, whose beautiful conjectures motivated my research on this subject and who, throughout my work, never ceased to give me advice and encouragement. This work might perhaps never have seen the light of day without the help of P. Deligne, who, over the course of long and numerous conversations, through his illuminating suggestions enabled me to smooth out a great many difficulties; I am happy to express here all the gratitude I owe him. Conversations with L. Breen, D. Quillen, and J.L. Verdier were very helpful to me; I extend my warm thanks to them. I also thank the participants in a seminar I gave on this subject at M.I.T. during the first semester of the 1970–71 academic year, whose questions, remarks, and suggestions were a valuable stimulus in writing this work.

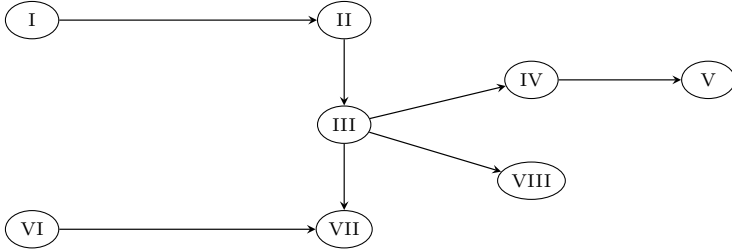
To Henri Cartan, who, after guiding my first steps in research and following my progress over the years, was kind enough to do me the honor of presiding over the committee to which I submitted this thesis, I express my profound gratitude. My gratitude also goes to Jean-Pierre Serre, who proposed a second subject to me, and to Michel Demazure, who agreed to serve on my committee.

Finally, I thank Madame Michon for the care she took in preparing the manuscript.

Prerequisites

We assume the reader to be familiar with the language of topoi, as presented in the first part of SGA 4 (especially Exposé IV), and with Verdier’s formalism of derived categories ([27], SGA 4 XVII). It is also advisable for the reader to have some knowledge of simplicial techniques ([10], [11], [23]), which play a fundamental role in our constructions; this knowledge is not, however, indispensable, since we recall in sufficient detail the results we need. Finally, we point out that, in order to approach this work, it is not necessary to have read the papers of André ([1], [2]), Grothendieck [17], and Quillen ([24], [25]), from which our work is, for the most part, logically independent.

Guide



Standard conventions

We shall sometimes use Deligne’s convention (SGA 4 XVII 0.1), consisting in denoting by the sign “=” a canonical functorial isomorphism.

When speaking of categories, topoi, “smallness,” etc., we shall sometimes omit to specify the ambient universe: it will be understood that this is a fixed universe of which $\mathbb{N}$ is an element.

We shall call a scheme (respectively, a separated scheme) what was formerly called a prescheme (respectively, a scheme).

CHAPTER I

RELATIVE HOMOTOPICAL ALGEBRA

1. Review.

1.1. Simplicial objects, homotopies ([10], [11], [23]).

Let Δ denote the category whose objects are the nonempty finite totally ordered sets and whose arrows are the order-preserving maps. For $n \in \mathbb{N}$ and $i \in [0, n]$, denote by

$$\delta_i^n \text{ (or } \delta_i) : [0, n-1] \longrightarrow [0, n]$$

the order-preserving injection that omits i , and by

$$\sigma_i^n \text{ (or } \sigma_i) : [0, n+1] \longrightarrow [0, n]$$

the order-preserving surjection that repeats i .

Let C be a category. A simplicial object of C is a functor from Δ° to C ; a morphism of simplicial objects of C is a morphism of the corresponding functors. If X is a simplicial object of C , for $n \in \mathbb{N}$ we write X_n in place of $X([0, n])$, and we sometimes denote simply by d_i^n (or d_i) the i -th face map $X(\delta_i^n)$ and by s_i^n (or s_i) the i -th degeneracy map $X(\sigma_i^n)$. We denote by

$$(1.1.1) \quad \text{Simpl}(C)$$

the category of simplicial objects of C . Analogously, for every integer $p \geq 1$ we define the category $p\text{-Simpl}(C)$ of p -simplicial objects of C , that is, functors from $(\Delta^p)^\circ (*)$ to C . Dually, we define the category $\text{Cosimpl}(C)$ of cosimplicial objects of C , that is, functors from Δ to C , and, for $p \geq 1$, the category $p\text{-Cosimpl}(C)$ of p -cosimplicial objects of C , that is, functors from $\Delta^p (*)$ to C . For every pair of integers $p, q \geq 1$, we further define the category $(p, q)\text{-Simpl}(C)$ of mixed simplicial objects in

$$(*) \quad \Delta^p = \underbrace{\Delta \times \cdots \times \Delta}_{p \text{ factors}}.$$

of type (p, q) in C , that is, functors from $(\Delta^p)^\circ \times \Delta^q$ to C . We shall mainly work with categories of the form $\text{Simpl}(C)$ or $2\text{-Simpl}(C)$. If X is a p -simplicial (resp. p -cosimplicial) object of C , we call the simplicial (resp. cosimplicial) object defined by $E \mapsto X(E, \dots, E)$ the diagonal subobject of X and denote it by ΔX .

For $X \in \text{ob } C$, let $K(X, 0)$ denote the constant functor $\Delta^\circ \rightarrow C$ with value X ; $K(X, 0)$ is also called the trivial simplicial object (or constant object) defined by X . The functor $X \mapsto K(X, 0)$ is left adjoint to the functor $Y \mapsto Y_0$: for $X \in \text{ob } C$ and $Y \in \text{ob } \text{Simpl}(C)$, the canonical restriction map

$$(1.1.2) \quad \text{Hom}(K(X, 0), Y) \rightarrow \text{Hom}(X, Y_0)$$

is an isomorphism. Since $K(X, 0)_0 = X$, the functor

$$K(-, 0) : C \rightarrow \text{Simpl}(C)$$

is therefore fully faithful, and its essential image consists of the simplicial objects all of whose face and degeneracy maps are isomorphisms. When no confusion can result, we shall simply write X instead of $K(X, 0)$.

Observe that if $i_0, i_1 : \{0\} \rightrightarrows E$ are arrows of Δ such that $i_0 \geq i_1$, there is a unique arrow $i : [0, 1] \rightarrow E$ of Δ such that $id^0 = i_0$ and $id^1 = i_1$. Hence, for $X \in \text{ob } C$ and $Z \in \text{ob } \text{Simpl}(C)$, the following diagram of sets, whose left-hand arrow is the natural restriction map, is exact:

$$(1.1.3) \quad \text{Hom}_{\text{Simpl}(C)}(Z, X) \longrightarrow \text{Hom}_C(Z_0, X) \begin{array}{c} \xrightarrow{\text{Hom}(d_0, X)} \\ \xrightarrow{\text{Hom}(d_1, X)} \end{array} \text{Hom}_C(Z_1, X)$$

An augmentation of Z to X is a morphism $f : Z \rightarrow X$ in $\text{Simpl}(C)$, i.e., by (1.1.3), a morphism $f_0 : Z_0 \rightarrow X$ such that $f_0 d_0 = f_0 d_1$. The datum of a simplicial object of C augmented to X (i.e. equipped with an augmentation to X) is equivalent to that of a simplicial object of C/X , or equivalently to that of a presheaf on the category of finite totally ordered sets with values

in C , taking the value X on the empty set.

For $Z \in \text{ob} \text{Simpl}(C)$, write $\pi_0(Z) \in \text{ob } C$ for the coequalizer of $(d_0, d_1) : Z_1 \rightrightarrows Z_0$, when the latter is representable; then (1.1.3) may be written as the isomorphism

$$(1.1.4) \quad \text{Hom}_{\text{Simpl}(C)}(Z, X) \xrightarrow{\sim} \text{Hom}_C(\pi_0(Z), X)$$

Thus, when coequalizers of pairs of arrows are representable in C , the functor $\pi_0(-)$ is left adjoint to the functor $K(-, 0)$, so that between the categories C and $\text{Simpl}(C)$ we have the following sequence of adjoint functors

$$(\pi_0(-), K(-, 0), (-)_0).$$

1.1.5. Let $u_0, u_1 : X \rightrightarrows Y$ be morphisms in $\text{Simpl}(C)$. A ([10]1.7) homotopy between u_0 and u_1 is the datum, for each morphism $f : E \longrightarrow [0, 1]$ in Δ , of a morphism $u_f : X(E) \longrightarrow Y(E)$ such that:

- (i) for every commutative triangle in Δ

$$\begin{array}{ccc} F & \xrightarrow{p} & E \\ & \searrow g & \swarrow f \\ & [0, 1] & \end{array}$$

the square

$$\begin{array}{ccc} X(E) & \xrightarrow{u_f} & Y(E) \\ \downarrow X(p) & & \downarrow Y(p) \\ X(F) & \xrightarrow{u_g} & Y(F) \end{array}$$

is commutative,

- (ii) $u_{(d^0 \varepsilon)} = u_0$, $u_{(d^1 \varepsilon)} = u_1$, where ε is the projection onto $\{0\}$.

If u is a homotopy between u_0 and u_1 , and $x : X' \longrightarrow X$ is a morphism in C , then the morphisms $u_f x(E) : X'(E) \longrightarrow Y(E)$, for $f : E \longrightarrow [0, 1]$, define a

homotopy between u_0x and u_1x ; similarly, if $y : Y \longrightarrow Y'$ is a morphism in C , the morphisms $yu_f : X(E) \longrightarrow Y'(E)$ define a homotopy between yu_0 and yu_1 . Likewise, if $S : C \longrightarrow C'$ is a functor, the morphisms $S(u_f) : S(X)(E) \longrightarrow S(Y)(E)$ define a homotopy between $S(u_0)$ and $S(u_1)$. We say that u_0 and u_1 are homotopic if they are congruent modulo the equivalence relation generated by the relation “there exists a homotopy between.” By the preceding remarks, the homotopy relation is compatible with composition of morphisms, so one can define a category

$$(1.1.5.1) \quad \text{Hotsimpl}(C)$$

having the same objects as $\text{Simpl}(C)$, and whose morphisms from X to Y are the homotopy classes of morphisms (in $\text{Simpl}(C)$) from X to Y . We say that $\text{Hotsimpl}(C)$ is the category of simplicial objects of C up to homotopy. There is a natural functor

$$\text{Simpl}(C) \longrightarrow \text{Hotsimpl}(C).$$

We say that a morphism in $\text{Simpl}(C)$ is a homotopy equivalence (or a homotopism) if its image in $\text{Hotsimpl}(C)$ is invertible. If $S : C \longrightarrow C'$ is a functor, S extends canonically to a functor, again denoted S , from $\text{Simpl}(C)$ to $\text{Simpl}(C')$, and, by the preceding remark, the latter extends canonically to a functor from $\text{Hotsimpl}(C)$ to $\text{Hotsimpl}(C')$.

The notion of homotopy between morphisms in $\text{Cosimpl}(C)$ and the category $\text{Hotcosimpl}(C)$ are defined analogously.

1.1.6. Let $p : E \longrightarrow B$ be a functor; denote by $\text{Simp}(p)$, or p , its natural extension to $\text{Simpl}(E)$. Let $u_0, u_1 : X \rightrightarrows Y$ be morphisms in $\text{Simpl}(E)$. We have seen that every homotopy u between u_0 and u_1 defines a homotopy $p(u)$ between $p(u_0)$ and $p(u_1)$.

Suppose that $p(u_0) = p(u_1) = f : S \longrightarrow T$. We call u a homotopy relative to f , or an f -homotopy, if $p(u)$ is the homotopy “constant with value f ,” i.e. $p(u)_s = f_a$ for every morphism $s : [a] \longrightarrow [0, 1]$ in Δ . If $S = T$ and $f = \text{Id}_S$, we say “ S -homotopy” instead of “ Id_S -homotopy.” We say that u_0 and u_1 are homotopic relative to f (or f -homotopic) if they are congruent modulo the equivalence relation

generated by the relation “there exists an f -homotopy between.” Let $v : X' \longrightarrow X$ be a morphism with projection $g : S' \longrightarrow S$. We have seen that a homotopy u between u_0 and u_1 defines a homotopy uv between u_0v and u_1v ; if u is an f -homotopy, then uv is an fg -homotopy. Likewise, if $w : Y \longrightarrow Y'$ is a morphism with projection $h : T \longrightarrow T'$, and if u is an f -homotopy between u_0 and u_1 , then wu is an hg -homotopy between wu_0 and wu_1 . It follows that, for $S \in \text{ob } \text{Simpl}(B)$, one can define a category

$$(1.1.6.1) \quad \text{Hotsimpl}(E)_S$$

having the same objects as $\text{Simpl}(E)_S$, and whose morphisms from X to Y are the S -homotopy classes of S -morphisms from X to Y . Suppose that the functor p is fibred (resp. cofibred). Then the same holds for $\text{Simpl}(p)$ (Giraud [12] 1.16).

In this case, giving an f -homotopy between two morphisms u_0 and u_1 over f is equivalent to giving an S -homotopy (resp. T -homotopy) between the corresponding morphisms $X \longrightarrow u_0^*(Y)$ and $X \longrightarrow u_1^*(Y)$ (resp. $u_{0*}X \longrightarrow Y$ and $u_{1*}X \longrightarrow Y$).

If $f = \text{Id}_S$ and u is an S -homotopy between u_0 and u_1 , then, for every morphism $g : S' \longrightarrow S$ (resp. $h : S \longrightarrow S''$), there is an S' -homotopy g^*u between g^*u_0 and g^*u_1 (resp. an S'' -homotopy h_*u between h_*u_0 and h_*u_1), defined by $(g^*u)_s = g^*(u_s)$ (resp. $(h_*u)_s = h_*(u_s)$) for $s : [a] \longrightarrow [0, 1] \in \text{Fl } \Delta$. Consequently, the functor $g^* : \text{Simpl}(E)_S \longrightarrow \text{Simpl}(E)_{S'}$ (resp. $h_* : \text{Simpl}(E)_S \longrightarrow \text{Simpl}(E)_{S''}$) extends to a functor

$$(1.1.6.2) \quad \begin{aligned} g^* : \text{Hotsimpl}(E)_S &\longrightarrow \text{Hotsimpl}(E)_{S'} \\ (\text{resp. } h_* : \text{Hotsimpl}(E)_S &\longrightarrow \text{Hotsimpl}(E)_{S''}), \end{aligned}$$

which permits one to define a category $\text{Hotsimpl}(E/B)$ fibred (resp. cofibred) over $\text{Simpl}(B)$ whose fibres are the categories $\text{Hotsimpl}(E)_S$.

The functor $p : E \longrightarrow B$ now being arbitrary again, let $q : F \longrightarrow B$ be a functor and $m : E \longrightarrow F$ a B -functor. Again denote by q (resp. m) the extension of q (resp. m) to $\text{Simpl}(F)$ (resp. $\text{Simpl}(E)$). Then m transforms an f -homotopy into an f -homotopy and hence, for $S \in \text{ob } \text{Simpl}(B)$, induces a functor